

From Sieve Words to Kneading Coordinates: A Rigorous Reformulation of the One-Dimensional Prime-Sieve Correspondence

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

We revisit the one-dimensional symbolic correspondence between cumulative Eratosthenes sieve words and the quadratic family $f_u(x) = 1 - ux^2$. The correspondence was originally formulated through a finite-horizon topological admissibility hypothesis. We separate three logically distinct issues: convergence in the ordered symbolic space, realizability as a kneading invariant, and topological conjugacy of dynamical systems. First, without any prime-gap hypothesis, we prove that the cumulative sieve words are strictly increasing in the parity-lexicographic order and converge in the product topology to RLR^∞ , with first disagreement exactly at the next prime. We identify the corresponding band-merging parameter as the unique root in $(1, 2)$ of $u^3 - 2u^2 + 2u - 2 = 0$, construct its three-state Markov partition, and obtain the exact entropy $\frac{1}{2} \log 2$ and even-return rigidity. We then prove that equality of the limiting sieve word with the kneading invariant does not yield a topological conjugacy: the natural shift orbit closure of RLR^∞ has three points and entropy zero, whereas the quadratic attractor has positive entropy.

At finite stages the universal admissibility hypothesis is false; we reproduce the explicit defect at sieve stage $k = 3$. A parity-gap lemma reduces every detected MSS defect to a prime gap of length at least $p_{k+1} - 1$. Combining this lemma with the Baker–Harman–Pintz theorem gives an unconditional admissible horizon of every length $H_k = p_{k+1}^{40/21-\varepsilon}$, for fixed $\varepsilon > 0$ and all sufficiently large k . Although current unconditional gap bounds do not reach the full physical horizon p_{k+1}^2 , we prove unconditionally that the density of MSS defects there is $O(p_{k+1}^{-1})$. The rigorous conclusion is therefore an ordered kneading-coordinate convergence and a parity factor, not an isomorphism between the prime sequence and a chaotic attractor. No statement about the infinitude of prime constellations is claimed.

Keywords: sieve of Eratosthenes; kneading theory; parity-lexicographic order; logistic map; prime gaps; symbolic dynamics; topological entropy.

MSC 2020: 37E05; 37B10; 11N35; 11A41.

1 Introduction

The Sieve of Eratosthenes is naturally expressed as a composition of periodic binary masks. In Wang [7], this elementary algebra was used to propose a symbolic correspondence with the quadratic family

$$f_u(x) = 1 - ux^2, \quad x \in [-1, 1], \quad 0 < u \leq 2, \quad (1)$$

at the first band-merging parameter $u_c \approx 1.543689$. The proposed correspondence has two attractive features. The cumulative sieve words move monotonically toward the word RLR^∞ ,

and the critical orbit of f_{u_c} has the same itinerary. A finite-horizon topological admissibility condition was introduced to interpret each sieve stage as a kneading coordinate.

Subsequent analysis found that the finite-stage condition is not universally valid: explicit Metropolis–Stein–Stein (MSS) defects occur at $k = 3$ and $k = 5$ [11]. This does not destroy every part of the construction. It instead forces a separation among three statements which are often conflated:

- (i) a sequence may converge to a kneading word in the product topology;
- (ii) a finite or infinite word may satisfy the MSS maximality inequalities and hence be realizable as kneading data;
- (iii) two dynamical systems may be topologically conjugate.

Statement (i) is purely combinatorial and, as we prove below, is unconditional. Statement (ii) is the genuine role of topological admissibility. Statement (iii) requires phase spaces, evolution maps, and a homeomorphism intertwining them; equality of two distinguished words is not sufficient.

The broader program has also led to a sequential ergodic formulation [9] and to numerical spectral models [8, 10]. The present paper deliberately isolates the topological layer. Its proofs do not use numerical orbit fitting, a non-autonomous density ansatz, or any conjecture concerning twin primes.

Main results

Our conclusions can be summarized as follows.

R1. The sieve words Q_k satisfy

$$Q_1 \prec Q_2 \prec \cdots \prec K_c = RLR^\infty, \quad d_{\vartheta}(Q_k, K_c) = \vartheta^{p_{k+1}},$$

for every standard symbolic metric d_{ϑ} , without assuming admissibility.

R2. The parameter u_c associated with RLR^∞ is algebraic. A three-state Markov partition gives

$$h_{\text{top}}(f_{u_c}) = \frac{1}{2} \log 2$$

and proves that successive visits to the L -region are separated by an even number of iterates.

R3. Under the natural shift-space interpretation, the limiting sieve system cannot be topologically conjugate to the quadratic attractor: their topological entropies are respectively 0 and $\frac{1}{2} \log 2$.

R4. A detected finite-horizon MSS defect forces a prime gap of length at least $p_{k+1} - 1$. The established bound $G(X) \ll X^{21/40}$ therefore gives unconditional horizon admissibility up to every length

$$H_k = p_{k+1}^{40/21-\varepsilon}$$

for fixed $\varepsilon > 0$ and all sufficiently large k .

R5. At the full physical horizon $N_k = p_{k+1}^2$, the number of detected defects satisfies

$$D_k \leq \frac{N_k}{p_{k+1} - 1} + 1, \quad \frac{D_k}{N_k - 1} = O(p_{k+1}^{-1}).$$

This defect-density result is unconditional and does not require a finite-defect conjecture.

These results retain a rigorous combinatorial skeleton of the original construction while placing a precise boundary around the word *isomorphism*. The finite sieve is an arithmetic mask, the limiting word is a kneading coordinate, and the quadratic system supplies a parity factor. None of these facts identifies the actual prime indicator with a typical quadratic orbit.

2 Sieve words and the unimodal order

2.1 Parity-lexicographic order

Let $\Sigma = \{L, R\}^{\mathbb{N}_0}$ and let $\sigma : \Sigma \rightarrow \Sigma$ be the left shift. We use the base order $L < R$.

Definition 2.1 (Parity-lexicographic order). Let $S = (s_n)$ and $T = (t_n)$ be distinct elements of Σ , and let

$$j = \min\{n \geq 0 : s_n \neq t_n\}.$$

Let r_j be the number of R symbols in the common prefix $s_0 \dots s_{j-1}$. If r_j is even, compare s_j and t_j using $L < R$; if r_j is odd, reverse the order. The resulting order is denoted by $\prec$.

This is the standard orientation-sensitive order for unimodal itineraries [5, 6]. For a one-sided word without a critical symbol, the MSS maximality condition is

$$\sigma^m(S) \preceq S \quad (m \geq 1). \quad (2)$$

We call a word satisfying (2) *self-admissible*. Endpoint conventions and completions matter when a finite word is promoted to a full kneading invariant; our finite-horizon definition below only records comparisons which are decided inside the stated horizon.

For $0 < \vartheta < 1$, define the standard symbolic metric

$$d_\vartheta(S, T) = \vartheta^{n(S, T)}, \quad n(S, T) = \min\{j \geq 0 : s_j \neq t_j\}, \quad (3)$$

with $d_\vartheta(S, S) = 0$.

2.2 Cumulative sieve masks

Let $2 = p_1 < p_2 < \dots$ be the primes and let

$$P_k = \prod_{j=1}^k p_j.$$

The k th cumulative sieve word $Q_k \in \Sigma$ is defined by

$$Q_k(n) = \begin{cases} L, & \gcd(n, P_k) = 1, \\ R, & \gcd(n, P_k) > 1, \end{cases} \quad n \geq 1, \quad Q_k(0) = R. \quad (4)$$

Thus Q_k is periodic with period P_k . Notice that the convention (4) marks each sieving prime p_j , $j \leq k$, by R along with its multiples. This convention is responsible both for the initial word $RLR\dots$ and for the pointwise limit established below.

Lemma 2.2 (Initial shield). *Let $p = p_{k+1}$. Then*

$$Q_k(0)Q_k(1)\dots Q_k(p) = RLR^{p-2}L. \quad (5)$$

Equivalently, $Q_k(n) = R$ for $2 \leq n < p$, while $Q_k(p) = L$.

Proof. Every prime smaller than $p = p_{k+1}$ is among $p_1, \dots, p_k$, and every composite $2 \leq n < p$ has a prime divisor smaller than p . Hence every $n \in [2, p)$ has nontrivial greatest common divisor with P_k . The next prime p is coprime to P_k . $\square$

Define

$$K_c = RLR^\infty. \quad (6)$$

Theorem 2.3 (Ordered symbolic convergence). *For every $k \geq 1$,*

$$Q_k \prec Q_{k+1} \prec K_c. \quad (7)$$

The first disagreement between Q_k and K_c occurs at index p_{k+1} , and therefore

$$d_{\vartheta}(Q_k, K_c) = \vartheta^{p_{k+1}}. \quad (8)$$

In particular $Q_k \rightarrow K_c$ in the product topology and pointwise on $\mathbb{N}_0$.

Proof. Put $p = p_{k+1}$. Adding the new sieve factor p does not alter any symbol with $1 \leq n < p$, while

$$Q_k(p) = L, \quad Q_{k+1}(p) = R.$$

By [lemma 2.2](#), the common prefix at indices $0, \dots, p-1$ is RLR^{p-2} . It contains

$$1 + (p-2) = p-1$$

symbols R . Since $p = p_{k+1}$ is odd for $k \geq 1$, this number is even. The comparison at index p therefore uses the base order $L < R$, proving $Q_k \prec Q_{k+1}$.

The same common prefix occurs when comparing Q_k with K_c , and at index p one again compares L in Q_k with R in K_c . Hence $Q_k \prec K_c$, the first-disagreement index is exactly p , and (8) follows from (3).

Finally, for each fixed $n \geq 2$, choose k large enough that a prime divisor of n belongs to $\{p_1, \dots, p_k\}$. Then $Q_k(n) = R$ for all subsequent stages. The symbols at 0 and 1 remain R and L , respectively, proving pointwise convergence to RLR^∞ . $\square$

Remark 2.4 (The order of limits). The limit in [theorem 2.3](#) is not the natural prime indicator. In the pointwise limit every fixed prime is eventually used as a sieve factor and hence is marked R . Thus K_c has exactly one L , at index 1. Retaining primes on a growing interval requires a diagonal choice of the sieve depth as the observation horizon grows. Pointwise convergence in k , diagonal sieving in the integer variable, and convergence in natural density are distinct operations.

3 The band-merging parameter and its exact Markov skeleton

We now identify the quadratic parameter associated with (6) without using any sieve admissibility assumption.

Proposition 3.1 (Algebraic band-merging parameter). *The polynomial*

$$F(u) = u^3 - 2u^2 + 2u - 2 \quad (9)$$

has a unique zero u_c in $(1, 2)$. Numerically,

$$u_c = 1.543689012692 \dots$$

If $b = u_c - 1$, then the critical orbit of f_{u_c} satisfies

$$0 \mapsto 1 \mapsto -b \mapsto b \mapsto b. \quad (10)$$

Consequently the itinerary of the critical value 1 under the partition $L = \{x < 0\}$, $R = \{x > 0\}$ is RLR^∞ .

Proof. We have $F(1) = -1$ and $F(2) = 2$. Moreover

$$F'(u) = 3u^2 - 4u + 2 = 3\left(u - \frac{2}{3}\right)^2 + \frac{2}{3} > 0,$$

so there is exactly one real zero in $(1, 2)$. Let $b = u_c - 1$. Equation $F(u_c) = 0$ is equivalent to

$$1 - u_c b^2 = b.$$

Therefore $f_{u_c}(1) = 1 - u_c = -b$ and

$$f_{u_c}(-b) = f_{u_c}(b) = 1 - u_c b^2 = b,$$

which proves (10) and its itinerary. $\square$

Theorem 3.2 (Markov partition, entropy, and even-return rigidity). *Let $b = u_c - 1$ and define*

$$J_L = [-b, 0], \quad J_0 = [0, b], \quad J_1 = [b, 1].$$

Up to their common endpoints,

$$f_{u_c}(J_L) = J_1, \quad f_{u_c}(J_0) = J_1, \quad f_{u_c}(J_1) = J_L \cup J_0. \quad (11)$$

The associated transition matrix is

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad \text{spec}(A) = \{0, \sqrt{2}, -\sqrt{2}\}. \quad (12)$$

Hence

$$h_{\text{top}}(f_{u_c}) = \log \rho(A) = \frac{1}{2} \log 2. \quad (13)$$

Furthermore, for every orbit which does not land on a partition endpoint, the time difference between two successive visits to J_L is even. In particular the word LL is forbidden.

Proof. On $[-b, 0]$ the map is increasing, with endpoint values $f_{u_c}(-b) = b$ and $f_{u_c}(0) = 1$. On $[0, b]$ it is decreasing with the same image $[b, 1]$. On $[b, 1]$ it is decreasing from b to $-b$. This proves (11) and (12). A direct determinant calculation gives

$$\det(\lambda I - A) = \lambda(\lambda^2 - 2),$$

so the spectral radius is $\sqrt{2}$. The standard entropy formula for a piecewise monotone Markov interval map yields (13); see, for example, de Melo and van Strien [3] and Lind and Marcus [4, Ch. 4].

For the return parity, put $C = [-b, b]$ and $H = [b, 1]$. The image relations above give

$$f_{u_c}(C) = H, \quad f_{u_c}(H) = C.$$

Thus an orbit alternates between the two sets C and H . Since $J_L \subset C$, all visits to J_L occur in the same parity class of the time variable. Successive visit times therefore differ by an even integer. In particular, a visit to J_L is followed by a visit to $H \subset \{x \geq 0\}$, excluding LL . $\square$

4 Kneading equality is not topological conjugacy

The phrase *topological isomorphism* requires two dynamical systems, not merely two distinguished symbolic words. The most direct shift-space realization of the limiting sieve word already gives a sharp obstruction.

Definition 4.1. Let

$$X_{K_c} = \overline{\{\sigma^n K_c : n \geq 0\}} \subset \Sigma$$

be the shift orbit closure of the limiting sieve word.

Theorem 4.2 (Entropy obstruction to conjugacy). *The shift system (X_{K_c}, σ) is not topologically conjugate to f_{u_c} on its invariant interval $[-b, 1]$.*

Proof. Since $K_c = RLR^\infty$,

$$X_{K_c} = \{RLR^\infty, LR^\infty, R^\infty\}.$$

It is a finite dynamical system and has topological entropy zero. By [theorem 3.2](#), the quadratic map has topological entropy $\frac{1}{2} \log 2 > 0$. Topological entropy is invariant under topological conjugacy, so no conjugacy exists. $\square$

Remark 4.3. The equality $\lim_k Q_k = K(f_{u_c})$ remains a correct statement about a coordinate in the ordered symbolic space. It says that the limiting word is the kneading invariant which determines the boundary of the admissible language at u_c . It does not say that the limiting word is a typical orbit, that its shift orbit closure is the quadratic attractor, or that the actual prime indicator has the same finite-word language as the quadratic system.

5 Finite-horizon admissibility and its failure at small stages

5.1 Detected defects

Let $H \geq 1$ and write $W_{k,H} = Q_k[0, H)$ for the prefix of length H .

Definition 5.1 (Detected MSS defect). A shift $m \in \{1, \dots, H-1\}$ is a *detected defect at horizon* H if there is a first index $j < H - m$ such that

$$Q_k(m+i) = Q_k(i) \quad (0 \leq i < j),$$

and the parity-lexicographic comparison at j gives

$$\sigma^m(Q_k) \succ Q_k.$$

Let $\mathcal{D}_k(H)$ be the set of detected defects, $D_k(H) = |\mathcal{D}_k(H)|$, and

$$\rho_k(H) = \frac{D_k(H)}{H-1}.$$

The prefix is *horizon-admissible* if $D_k(H) = 0$.

This convention makes no assertion when a comparison is not decided before the suffix exits the horizon. It is exactly the finite information available in the sieve-valid window.

Proposition 5.2 (Explicit failure at $k = 3$). *At stage $k = 3$, with sieve primes $2, 3, 5$ and physical horizon $H = p_4^2 = 49$, the shift $m = 22$ is a detected MSS defect.*

Proof. The first ten symbols of Q_3 and its shift are

i	0	1	2	3	4	5	6	7	8	9
$Q_3(i)$	R	L	R	R	R	R	R	L	R	R
$Q_3(22+i)$	R	L	R	R	R	R	R	L	R	L

The common prefix of length nine contains seven R symbols, so the order is reversed at the first disagreement. At position 9, Q_3 has R and the shift has L ; under the reversed order $R < L$. Thus $Q_3 \prec \sigma^{22}Q_3$, and the disagreement is detected at the integer $22 + 9 = 31 < 49$. $\square$

Corollary 5.3. *The assertion that every cumulative sieve stage is kneading-admissible on its entire physical horizon is false.*

The analogous defect at $k = 5$ is generated by the prime gap 113–127; see Wang [11]. We now prove the structural reduction which explains why such defects are rare.

6 The parity-gap reduction

For a prime q , let q^+ denote the next prime and write $g(q) = q^+ - q$. Define

$$G(X) = \max\{g(q) : q \leq X, q \text{ prime}\}. \quad (14)$$

Lemma 6.1 (Parity-gap lemma). *Let $p = p_{k+1}$ and let $p < H \leq p^2$. If $m \in \mathcal{D}_k(H)$, then*

- (i) $q = m + 1$ is prime;
- (ii) $g(q) \geq p - 1$.

Proof. By lemma 2.2, the reference prefix begins

$$Q_k = RLR^{p-2}L \dots$$

We compare $\sigma^m Q_k$ with Q_k .

At position 0, if $Q_k(m) = L$, then the shift begins with L while Q_k begins with R . The common prefix is empty, so the base order applies and the shift is smaller. A defective shift must therefore satisfy $Q_k(m) = R$. The common prefix now contains one R , and the order is reversed at the next disagreement.

At position 1, the reference word has $Q_k(1) = L$. If $Q_k(m+1) = R$, then under the reversed order the shifted R is smaller than the reference L , again defeating the shift. Hence a defect requires $Q_k(m+1) = L$. Since $2 \leq m+1 < H \leq p^2$, every composite integer in this range has a prime factor strictly smaller than p and is marked R by Q_k . Therefore $q = m+1$ is prime. In particular $q \geq p$ and is odd.

Let $r < H - m$ be the first disagreement which detects the defect. We claim that $r \geq p$. Otherwise $2 \leq r \leq p-1$, the reference symbol is $Q_k(r) = R$, and the shifted symbol must be $Q_k(m+r) = L$. By the same sieve-validity argument, $m+r$ is prime. Both $m+1$ and $m+r$ are odd primes, so their difference $r-1$ is even and hence r is odd. The common reference prefix through position $r-1$ contains

$$1 + (r-2) = r-1$$

symbols R , an even number. The order at position r is therefore unflipped, and the shifted symbol L is smaller than the reference symbol R , contradicting that r detects a defect. Hence $r \geq p$.

The two words therefore agree at every position $2 \leq j \leq p-1$, so $Q_k(m+j) = R$ there. Since $m+j > q \geq p$, each such integer is divisible by one of the sieving primes and is composite. Equivalently, the integers

$$q+1, q+2, \dots, q+p-2$$

are all composite. The next prime satisfies $q^+ \geq q+p-1$, and hence $g(q) \geq p-1$. $\square$

Corollary 6.2 (Prime-gap sufficient condition). *Let $p = p_{k+1}$ and $p < H \leq p^2$. If*

$$G(H) < p-1, \tag{15}$$

then $W_{k,H}$ is horizon-admissible.

Proof. The contrapositive of [lemma 6.1](#) states that a defect would produce a prime $q \leq H$ with $g(q) \geq p-1$, contradicting (15). $\square$

Condition (15) is sufficient, not necessary. In particular a long prime gap does not by itself force the parity of every later tie-breaker to create an MSS defect.

7 An unconditional long admissible horizon

The unconditional all- x consecutive-prime estimate needed here is due to Baker et al. [1]:

$$p_{n+1} - p_n \ll p_n^{21/40}, \tag{16}$$

for all sufficiently large n . In particular, $G(X) \ll X^{21/40}$.

Theorem 7.1 (BHP admissible-prefix theorem). *Fix*

$$0 < \varepsilon < \frac{19}{21}$$

and put

$$H_k = \left\lfloor p_{k+1}^{40/21-\varepsilon} \right\rfloor. \tag{17}$$

Then W_{k,H_k} is horizon-admissible for all sufficiently large k .

Proof. Let $p = p_{k+1}$. Since $1 < 40/21 - \varepsilon < 2$, we have $p < H_k < p^2$ for all sufficiently large p . By (16),

$$G(H_k) \ll H_k^{21/40} \ll p^{(40/21-\varepsilon)(21/40)} = p^{1-(21/40)\varepsilon} = o(p).$$

Thus $G(H_k) < p-1$ for sufficiently large k , and [corollary 6.2](#) applies. $\square$

Corollary 7.2 (Exponent criterion). *Suppose an all- X prime-gap theorem gives $G(X) \ll X^\theta$. Then every horizon $H_k = p_{k+1}^a$ with*

$$1 < a < \min\{2, 1/\theta\}$$

is eventually horizon-admissible through [corollary 6.2](#). The full quadratic horizon $H_k = p_{k+1}^2$ follows from this route if $G(X) = o(X^{1/2})$.

Remark 7.3 (Why the full horizon remains open by this method). At $X = p^2$, the Baker–Harman–Pintz exponent gives only

$$G(p^2) \ll p^{21/20},$$

which is larger than the required shield $p-1$. A Riemann-hypothesis bound of order $O(X^{1/2} \log X)$ also does not imply (15) at $X = p^2$. In contrast, Cramér-type polylogarithmic bounds [2] would imply eventual full-horizon admissibility. The latter remains a conditional statement.

8 Unconditional decay of the defect density

Full admissibility is much stronger than the vanishing of the proportion of bad shifts. The parity-gap lemma proves the latter unconditionally.

Theorem 8.1 (Defect-count bound). *Let $p = p_{k+1}$ and $p < H \leq p^2$. Then*

$$D_k(H) \leq \frac{H}{p-1} + 1. \quad (18)$$

Consequently, at the physical horizon $N_k = p_{k+1}^2$,

$$\rho_k(N_k) \leq \frac{1}{p_{k+1}-1} + O(p_{k+1}^{-2}) \longrightarrow 0. \quad (19)$$

Proof. By lemma 6.1, every defect m determines the prime $q = m + 1$ and a consecutive-prime interval $[q, q^+)$ of length at least $p - 1$. The map $m \mapsto q$ is injective, and consecutive-prime intervals associated with distinct q are disjoint.

All their left endpoints satisfy $q \leq H$. At most one of these disjoint intervals can cross the endpoint H . The remaining intervals lie inside $[0, H]$, have total length at most H , and each has length at least $p - 1$. Hence there are at most $H/(p - 1)$ of them, plus at most one crossing interval. This proves (18). Dividing by $H - 1$ and setting $H = p^2$ gives (19). $\square$

Remark 8.2 (What defect density does not prove). The limit $\rho_k(N_k) \rightarrow 0$ proves asymptotic admissibility in the defect-density sense. It does not by itself construct a true admissible word at Hamming distance $o(N_k)$ from W_{k,N_k} , nor does it prove shadowing by a quadratic orbit. A sparse set of failed maximality comparisons is not the same object as a sparse set of symbol edits. Any passage from (19) to equality of ergodic cylinder statistics requires an additional repair or shadowing theorem.

9 An alternative exact prime word and its tradeoff

The obstruction above is partly caused by marking each sieving prime itself as R . If one instead uses the natural prime word, absolute self-admissibility becomes elementary, but the band-merging limit is lost.

Definition 9.1. Define $\mathcal{P} \in \Sigma$ by

$$\mathcal{P}(0) = R, \quad \mathcal{P}(1) = L,$$

and, for $n \geq 2$,

$$\mathcal{P}(n) = \begin{cases} L, & n \text{ is prime,} \\ R, & n \text{ is composite.} \end{cases}$$

Proposition 9.2 (Self-admissibility of the natural prime word). *For every $m \geq 1$,*

$$\sigma^m \mathcal{P} \prec \mathcal{P}.$$

Proof. If $\mathcal{P}(m) = L$, the comparison is decided at position 0: the shift begins with L , while $\mathcal{P}$ begins with R , so the shift is smaller in the base order.

Suppose $\mathcal{P}(m) = R$. Then $m \geq 4$ is composite and both words begin with R , so the order is reversed at the next disagreement. If $\mathcal{P}(m+1) = R$, position 1 compares the shifted R with $\mathcal{P}(1) = L$; under the reversed order the shift is smaller. If $\mathcal{P}(m+1) = L$, then $m+1$ is an odd prime at least 5, so $m+2$ is even and composite. The words agree as RL at positions 0, 1, the prefix still contains one R , and at position 2 the shifted R is smaller than $\mathcal{P}(2) = L$ in the reversed order. Thus every nonzero shift is defeated by position 2 at the latest. $\square$

Remark 9.3. The proposition gives an unconditional inverse-kneading encoding of the prime indicator. However, $\mathcal{P}$ begins $RLLL\cdots$, not RLR^∞ , and therefore does not select the band-merging parameter u_c . Exact prime coding and the particular one-dimensional band-merging skeleton are different constructions.

10 Interpretation and scope

The preceding results suggest the following replacement for the original strong isomorphism statement.

The cumulative sieve masks form a strictly ordered path in the full symbolic space and converge to the band-merging kneading coordinate RLR^∞ . The quadratic map at that coordinate realizes the parity factor of the sieve grammar. Finite-stage realization as kneading data is valid on an unconditional growing subquadratic horizon and has vanishing defect density on the full quadratic horizon.

This statement retains the exact algebraic and topological facts while avoiding four overextensions.

- (1) **A finite prefix is not a unique parameter.** Even when every comparison visible in a finite word is admissible, a completion rule is needed before the word becomes an infinite kneading invariant. Different completions can determine an interval of parameters rather than a single u_k .
- (2) **Pointwise sieve convergence erases the primes.** The word RLR^∞ is a boundary coordinate, not the natural prime sequence; see [remark 2.4](#).
- (3) **A kneading invariant is not a typical orbit.** It controls the language of a unimodal system. Equality of a limiting mask with a kneading word does not supply a conjugating homeomorphism; [theorem 4.2](#) rules out the natural shift-space version.
- (4) **Asymptotic admissibility is not arithmetic shadowing.** The defect-density theorem is a genuine unconditional macroscopic statement, but the stronger orbit-shadowing problem remains separate; see [remark 8.2](#).

The numerical computations in Wang [7, 11] remain useful in their proper role: they discovered the small-stage defects, quantified their scarcity over a large finite range, and motivated [lemma 6.1](#). The proofs in the present paper neither infer an infinite theorem from that finite range nor discard the numerical observations. They identify exactly which conclusions are stable under the transition from computation to proof.

11 Conclusion

The topological admissibility condition is essential for assigning a sieve word itself as kneading data, but it is not needed for the ordered convergence

$$Q_k \nearrow RLR^\infty$$

or for the algebraic identification of the band-merging parameter. The universal finite-stage condition is false, while two rigorous replacements are available: an unconditional admissible prefix of length $p_{k+1}^{40/21-\varepsilon}$ and an unconditional $O(p_{k+1}^{-1})$ defect density on the full physical horizon.

At the same time, the entropy obstruction shows that the surviving result cannot be called a topological conjugacy between the limiting sieve word and the chaotic quadratic attractor.

The correct object is a kneading-coordinate convergence together with a parity factor. This reformulation provides a stable base for subsequent spectral and non-autonomous work: stronger arithmetic conclusions must enter through a separately stated shadowing, shrinking-target, or sieve-remainder theorem, not through kneading equality alone.

Data and code availability

All statements in this paper are proved analytically and require no numerical data. Code and data associated with the motivating computations are available from the repositories cited in Wang [7, 11] and from the author upon reasonable request.

Acknowledgements

The author thanks the reviewers of the related manuscripts for comments that motivated a sharper separation between symbolic convergence, admissibility, and conjugacy. AI-assisted tools were used for auxiliary typesetting and proof auditing; all mathematical statements and conclusions are the responsibility of the author.

Disclosure statement

The author reports no competing interests.

References

- [1] R. C. Baker, G. Harman, and J. Pintz. The difference between consecutive primes, II. *Proceedings of the London Mathematical Society*, 83(3):532–562, 2001. doi: 10.1112/plms/83.3.532.
- [2] Harald Cramér. On the order of magnitude of the difference between consecutive prime numbers. *Acta Arithmetica*, 2:23–46, 1936.
- [3] Welington de Melo and Sebastian van Strien. *One-Dimensional Dynamics*, volume 25 of *Ergebnisse der Mathematik und ihrer Grenzgebiete*. Springer, Berlin, 1993. doi: 10.1007/978-3-642-78043-1.
- [4] Douglas Lind and Brian Marcus. *An Introduction to Symbolic Dynamics and Coding*. Cambridge University Press, Cambridge, 1995. doi: 10.1017/CBO9780511626302.
- [5] Nicholas Metropolis, Myron L. Stein, and Paul R. Stein. On finite limit sets for transformations on the unit interval. *Journal of Combinatorial Theory, Series A*, 15(1):25–44, 1973. doi: 10.1016/0097-3165(73)90033-2.
- [6] John Milnor and William Thurston. On iterated maps of the interval. In James C. Alexander, editor, *Dynamical Systems*, volume 1342 of *Lecture Notes in Mathematics*, pages 465–563. Springer, Berlin, 1988. doi: 10.1007/BFb0082847.
- [7] Liang Wang. The emergence of prime distribution from low-dimensional deterministic chaos. *Research in Mathematics*, 13(1):2684334, 2026. doi: 10.1080/27684830.2026.2684334.
- [8] Liang Wang. An area-preserving Hénon-map model for the Riemann zeros: A deterministic-dynamics approach with quantum and dissipative spectral evidence. Manuscript under review, 2026.

- [9] Liang Wang. A sequential Birkhoff theorem for slow logarithmic drift in non-uniformly expanding unimodal maps. Manuscript under review, 2026.
- [10] Liang Wang. Spectral isomorphism between renormalization flow in non-autonomous quadratic maps and Riemann zeros. Manuscript under review, 2026.
- [11] Liang Wang. Transient chaos and topological bounds in prime dynamics: Revisiting the one-dimensional sieve mapping. Manuscript under review, 2026.